

Architecture: Enterprise Data Platform

Meridian Commercial Bank - Internal Document

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Overview

The Enterprise Data Platform (EDP) ingests, stores, and serves data from across Meridian Commercial Bank's operational systems for analytics, reporting, regulatory compliance, and (as of 2026) the internal AI Assistant's knowledge retrieval layer.

Components

- Ingestion Layer: Consumes events from the Core Banking Integration Layer's Event Publisher, plus scheduled extracts from operational databases and uploaded document repositories (policies, runbooks, incident reports, etc.).
- Object Storage: Raw and processed data lake, partitioned by domain (payments, retail_banking, core_banking, security) and date.
- Document Index: A hybrid dense/sparse vector index (embeddings + keyword/BM25) over unstructured documents, with metadata filtering by department, document_type, access_level, and created_date. This is the retrieval backend for the internal AI Assistant.
- Warehouse: Structured analytics tables for BI dashboards and regulatory reporting, refreshed nightly.

Data Governance

- All documents ingested into the Document Index carry an access_level tag (public, internal, confidential, restricted). The AI Assistant's retrieval layer enforces role-based filtering on this field so that, for example, a Viewer-role user cannot retrieve confidential documents even if the document is topically relevant to their query.
- Data lineage is tracked per ingestion job; every record in the Warehouse can be traced back to its source system and ingestion timestamp.

Access Model

Access to raw Object Storage is restricted to Data Platform engineers. Access to the Warehouse and Document Index is mediated entirely through service APIs (including the AI Assistant backend), which enforce authentication and role-based authorization before any query executes.

Related Documents

- ARCH-identity-access-management (authentication/authorization enforced at the API layer described above)
- ARCH-core-banking-integration (primary event source for the Ingestion Layer)